

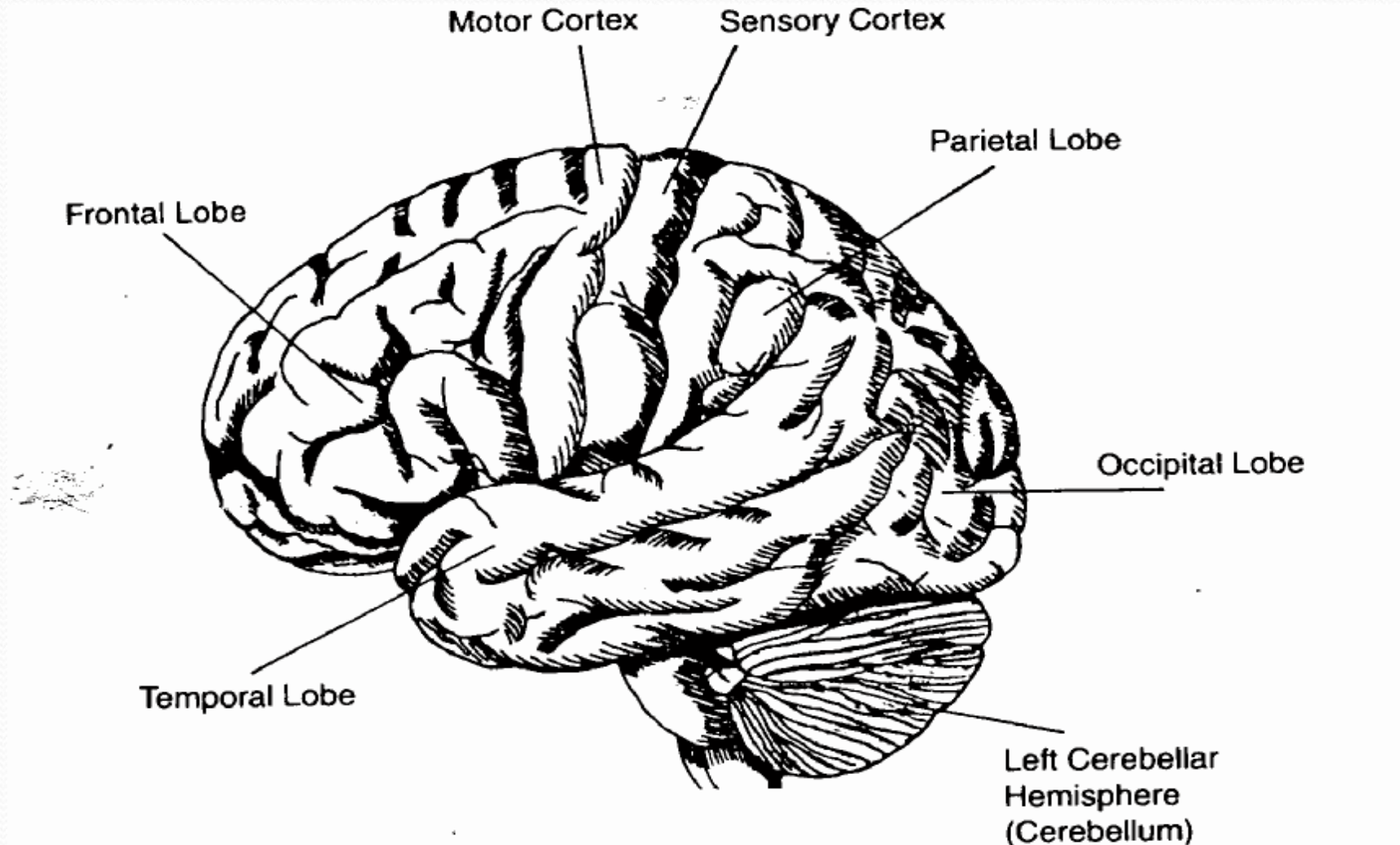
Traumatic Brain Injury

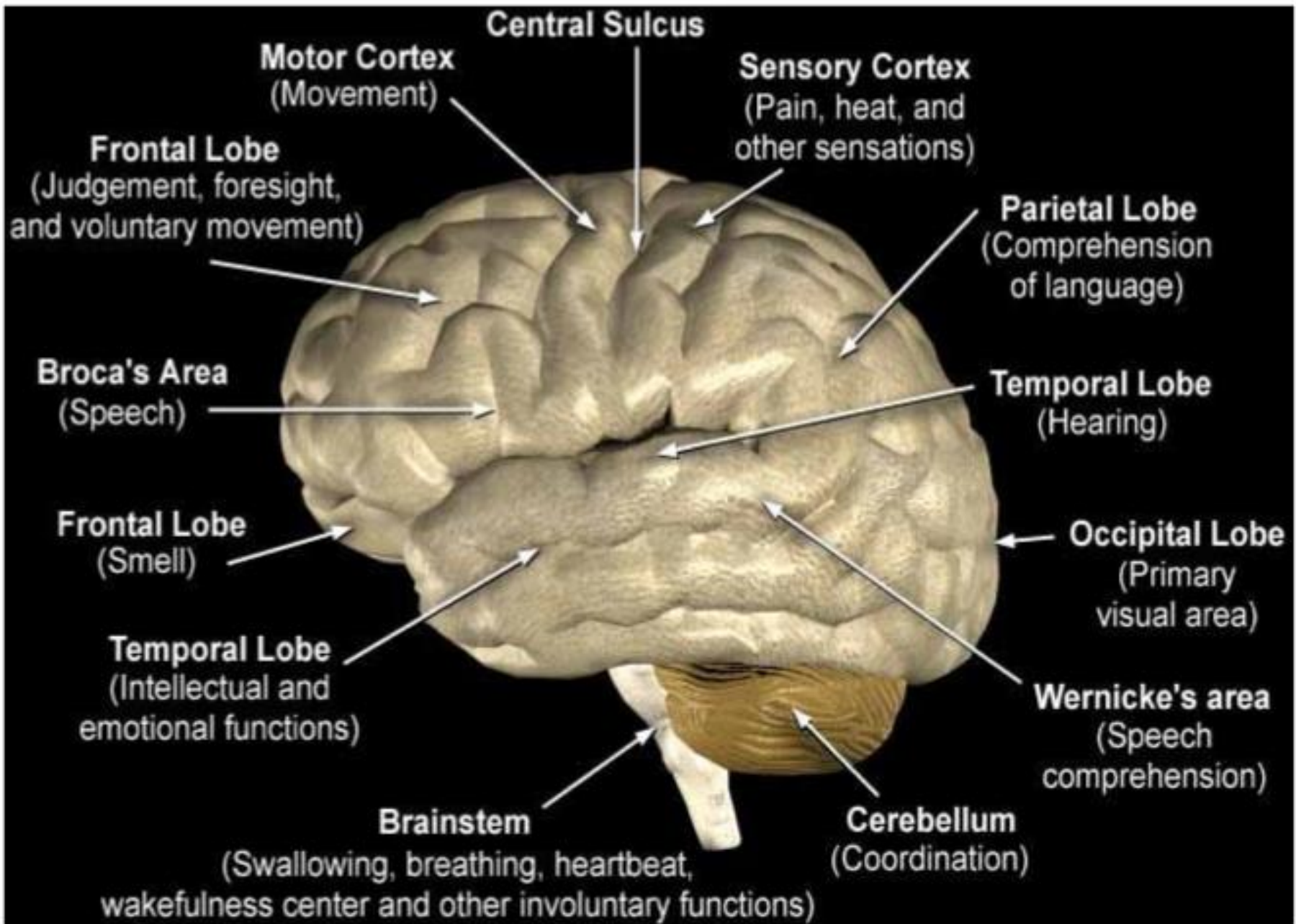
Dr Trevor Powell
Consultant Clinical Neuropsychologist
Berkshire Healthcare NHS Foundation
Trust
Chairman Headway Thames Valley

Content of talk

- What is a TBI? Brief understanding the brain and levels of severity of injury.
- Effects of TBI and the 'hidden disabilities'
- The role of the Neuropsychologist in assessment – including symptom validity testing
- The long term - Posttraumatic growth studies
- Questions and discussion

The Brain: 1million synaptic connections every second: 2 decades to develop: 10 lbs: Fronto-temporal areas injured in TBI - blancmange





What is a head injury

- Definition

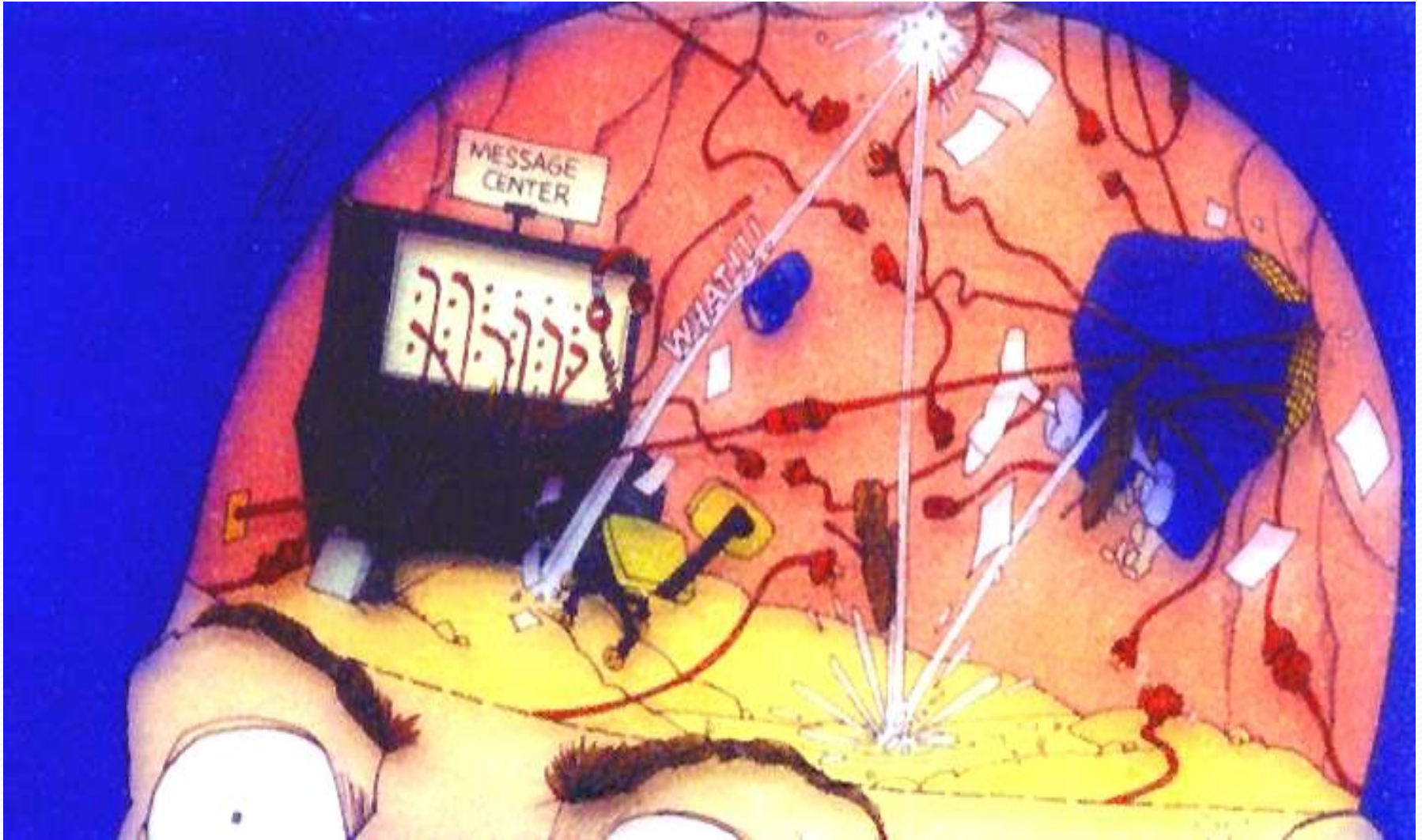
- “TBI is a non-degenerative, non-congenital **insult** to the **brain** from an **external** mechanical force, usually due to rapid acceleration or deceleration, possibly leading to temporary or permanent impairment of **cognitive, physical and psychosocial functions** with an associated diminished or altered state of consciousness.”

Neuro-anatomy of TBI

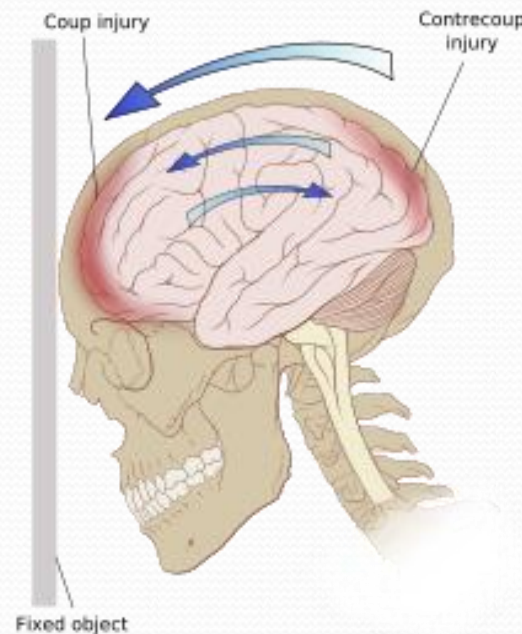


How the brain works...

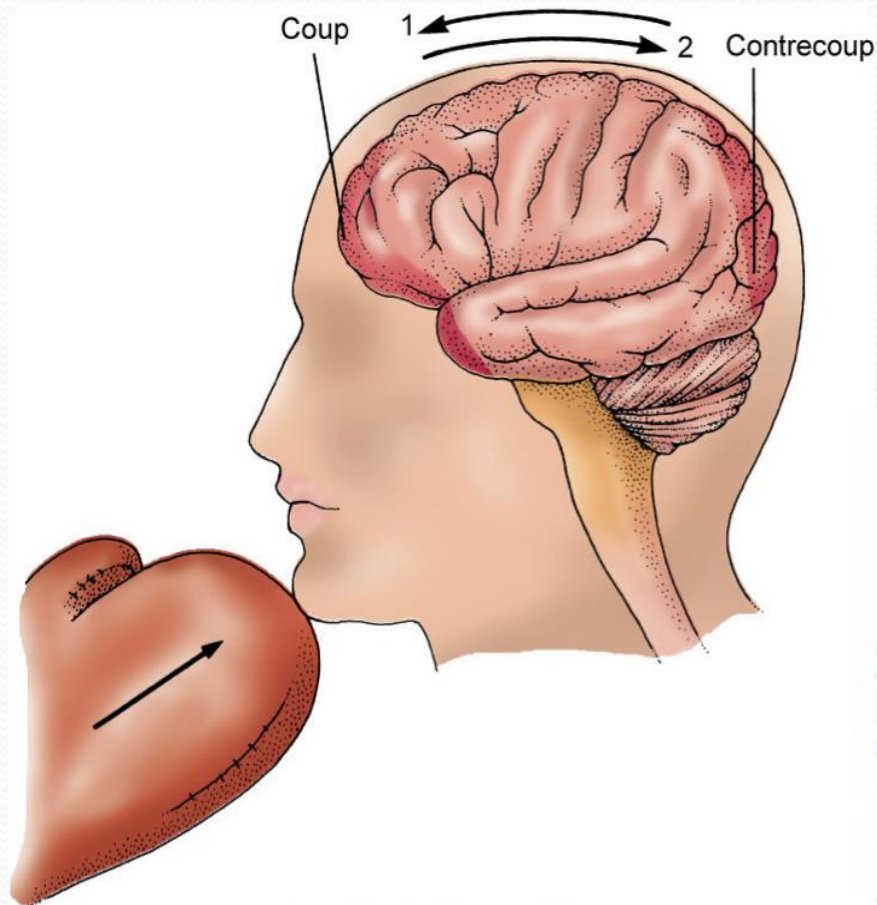
How the brain might look after a TBI...



Coup Contre-Coup Injury: describes contusions (bruises) that are both at the site of impact and on the opposite side of the brain.

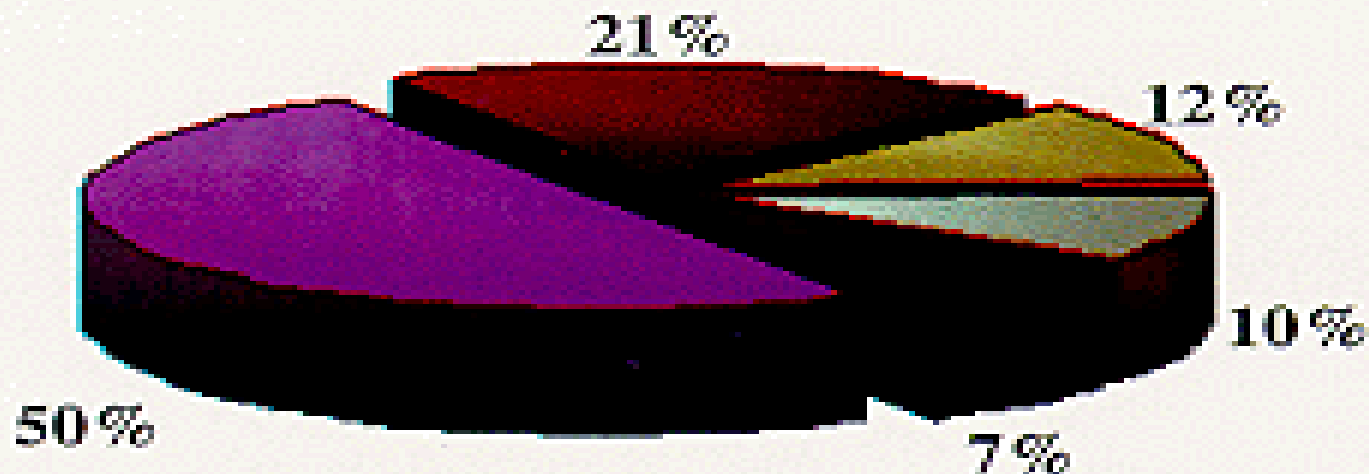


1. Pathology



Causes of Head Injury (TBI)

- Motor vehicle accidents-50 %
- Falls-21 %
- Assaults/acts of violence-12 %
- Sports & recreation-10 %
- Other-7 %



High Risk Groups

- Young men and elderly
- Previous head injuries
- Residents of inner cities
- Alcohol and drug abuse
- Low-income

Head Injury: The facts and figures

- Incidence: > 1 m people in UK attend Hospital with HI
- Approximately 45 -50 per 100,000 left with some residual disability
- Prevalence: 150 disabled survivors per 100,000 of population
- Males 2-3 times more likely – high risk group 15 - 29 and over 65

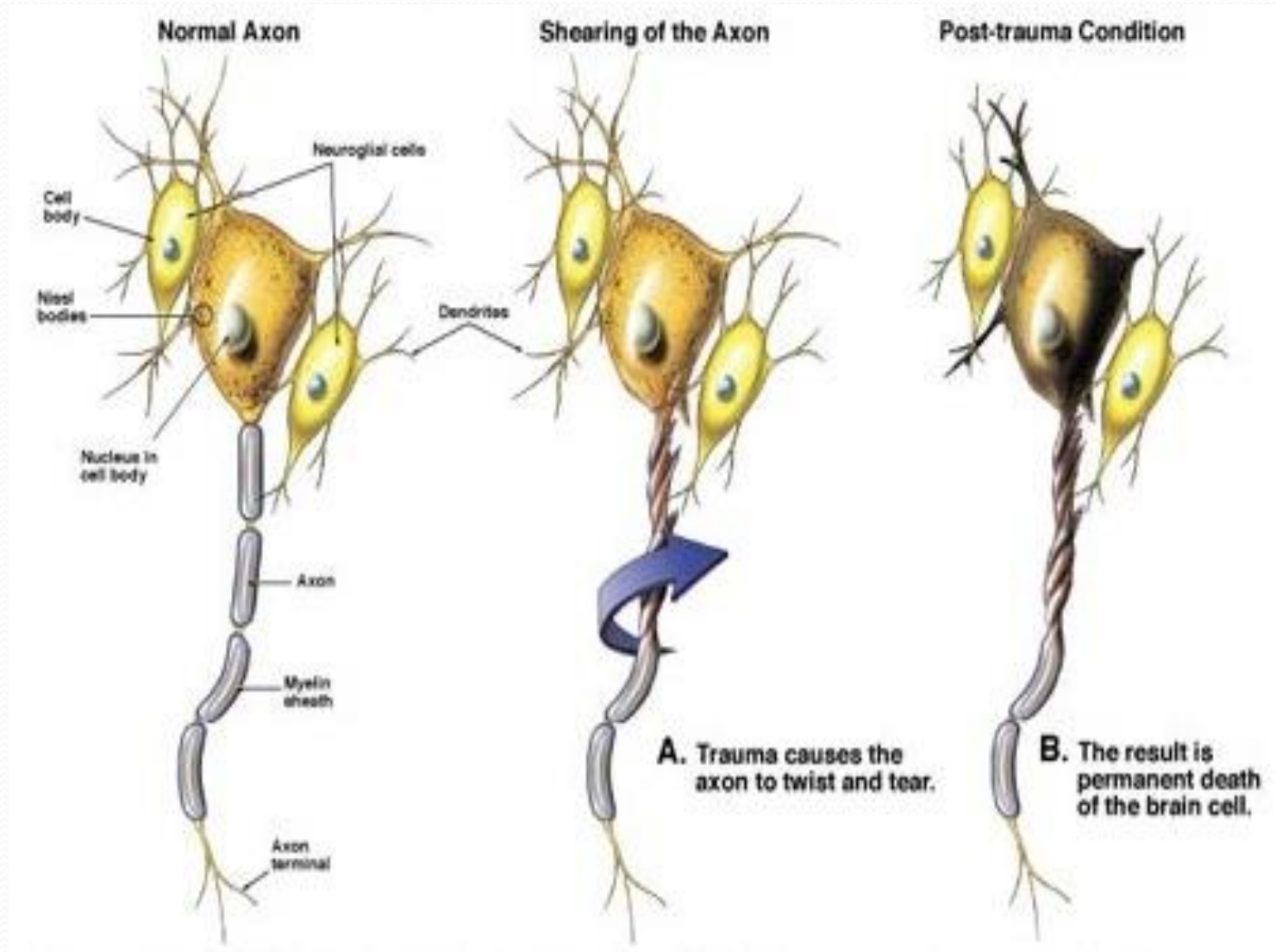
How is the brain damaged?

- Depressed Skull Fracture (the broken bone presses inwards)
- Laceration (tearing of brain tissue)
- Contusion (bruising of the brain)
- Haemorrhage (bleeding inside the skull)
- Diffuse axonal injury (microscopic nerve damage)
- Anoxia (lack of oxygen to the brain)

Mechanisms of Injury

- Assault
 - Falls
 - RTAs
 - Sports
-
- Over half involve alcohol!!

Diffuse axonal injury



Haematoma

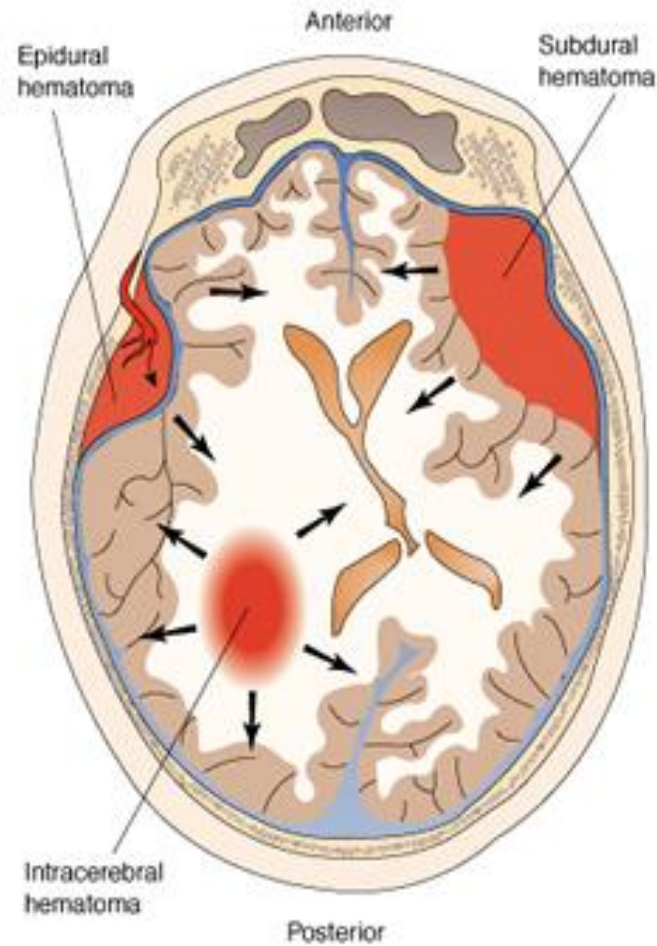
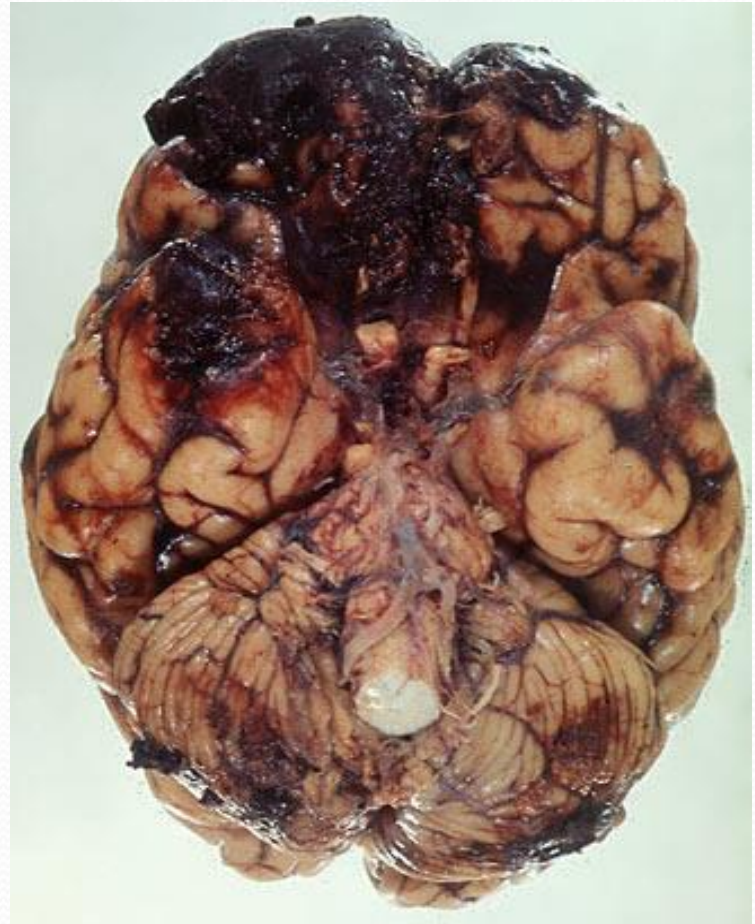
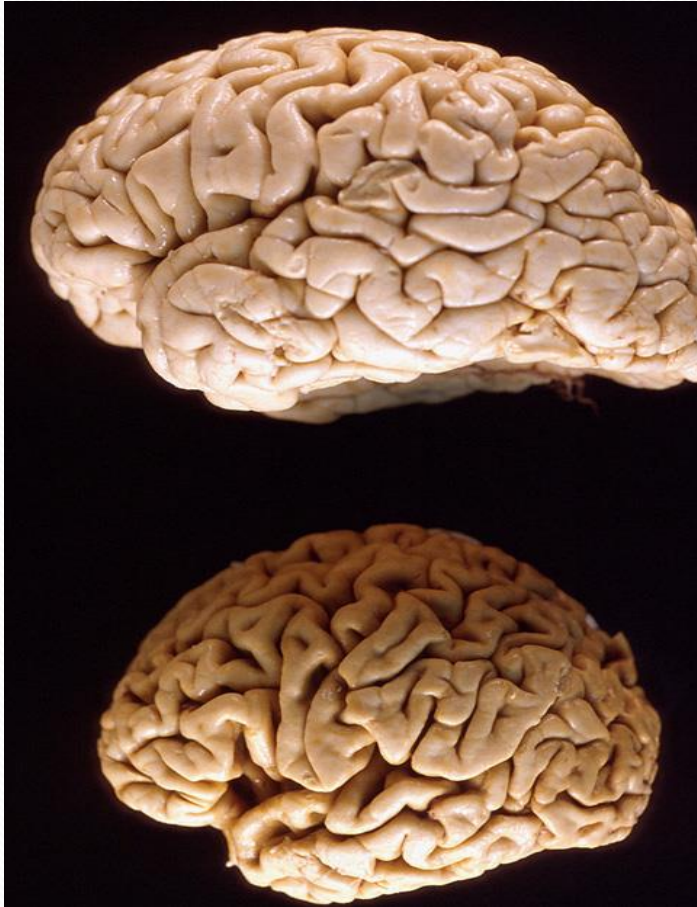
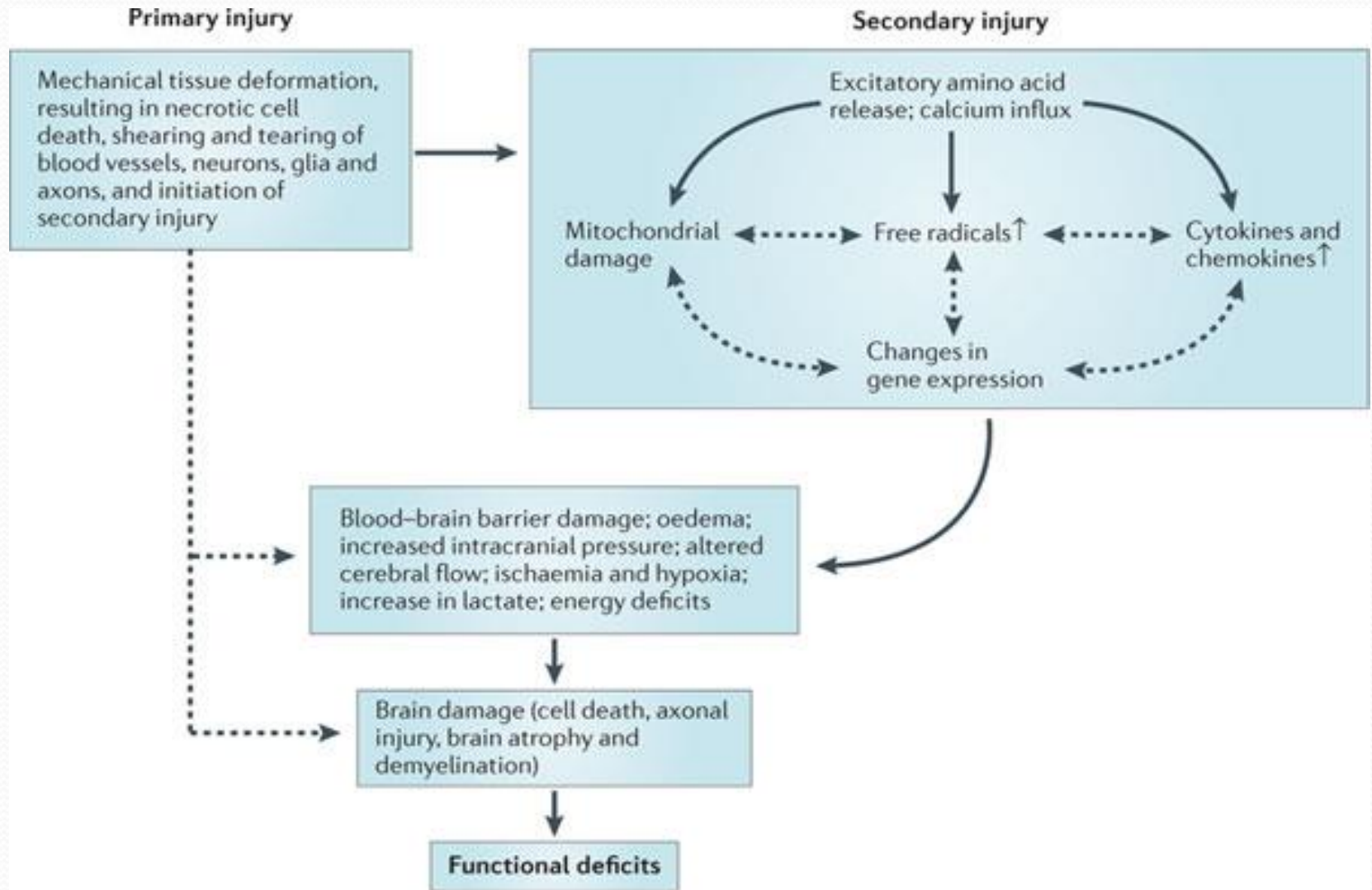


Figure 52-8 Location of epidural, subdural, and intracerebral hematomas.

Cerebral contusion (severe)



Secondary damage



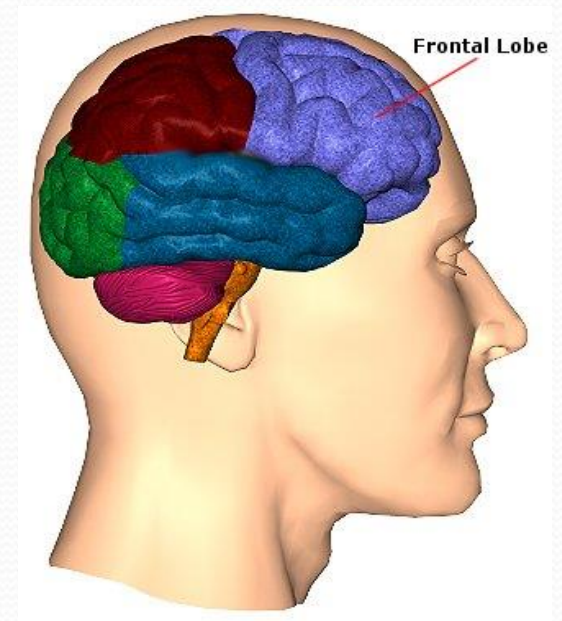
Case 4: James Cracknell

- <http://www.youtube.com/watch?v=gIWj8g88Zaw>
- Checked Nov 14

Frontal Lobes

Involved in:

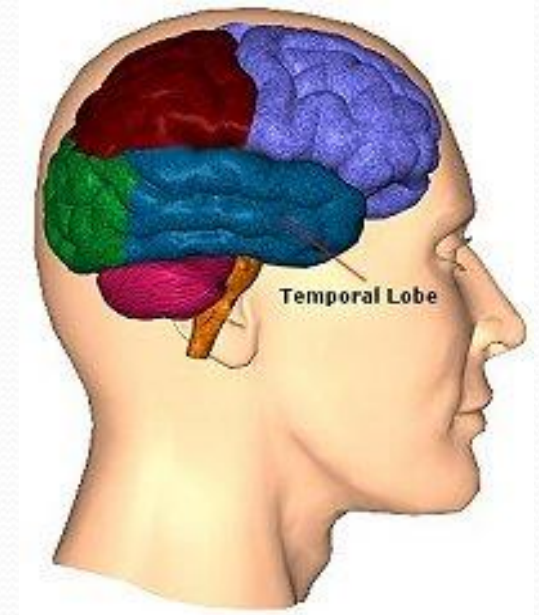
- Awareness of self and changes in your abilities
- Understanding and controlling emotions
- Anticipation of problems/events
- Planning and organising events
- Concentration and attention
- Production of speech
- Initiating activities
- Movement



Temporal Lobes

Involved in:

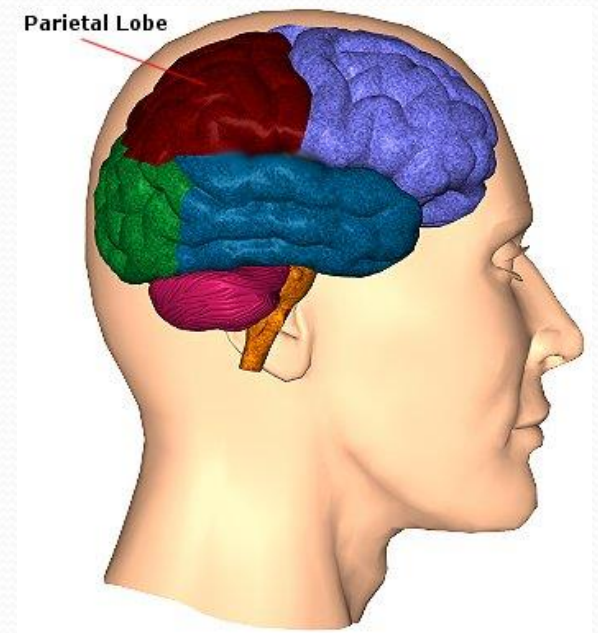
- Understanding speech sounds
- Verbal and non-verbal memory
- Interpretation of auditory information
- Reading comprehension and retention



Parietal Lobes

Involved in:

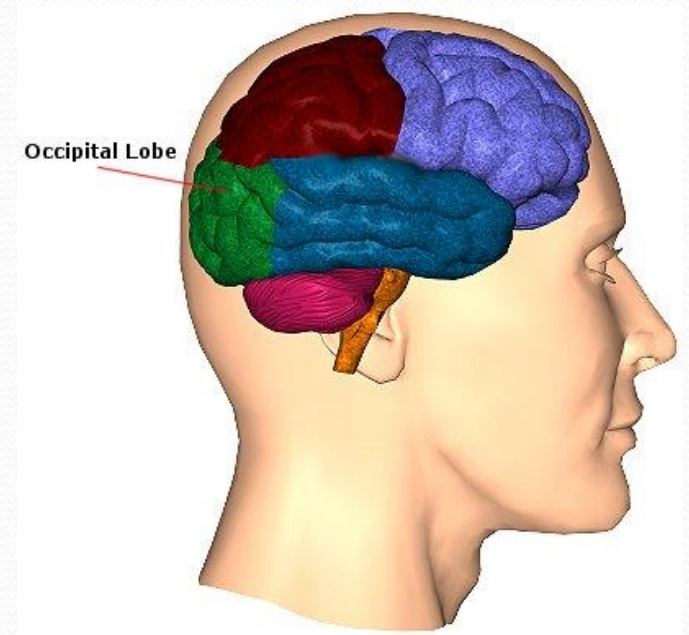
- Spatial awareness
- Attention to items in space around you (e.g. noticing things on the left and right)
- Recognising bodily sensations
- Writing
- Maths



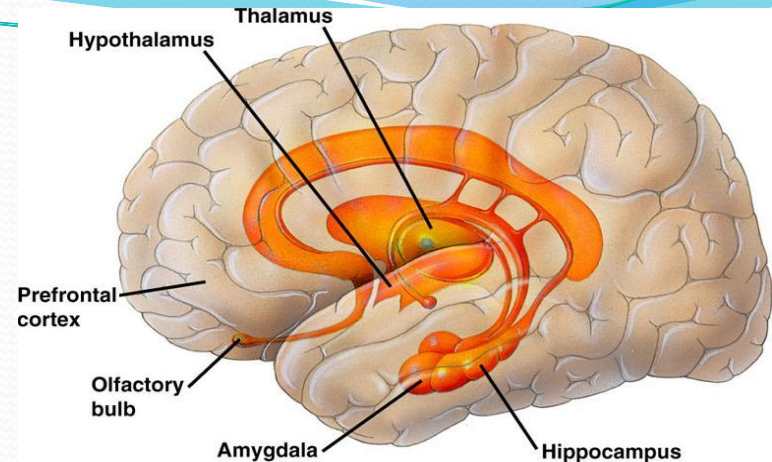
Occipital Lobes

Involved in:

- Vision
- Recognising objects



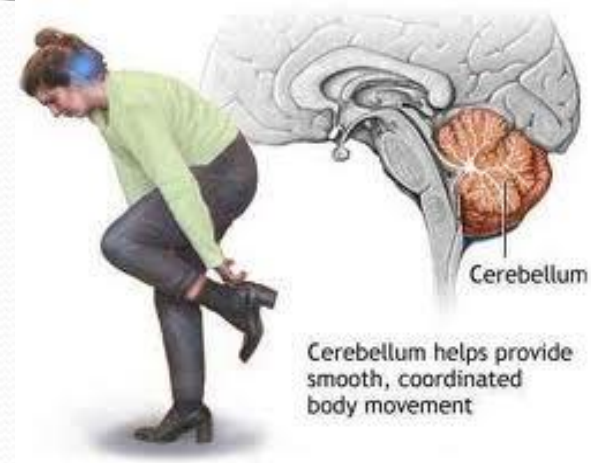
Limbic System



- Often referred to as the “emotional brain” and the amygdala the “emotional hub”
- Involved in the regulation of motivations and emotions, particularly those related to survival (e.g., fear, anger, sexual behaviour).
- Also involved in the formation and retrieval of long-term memory (role of the hippocampus)

Cerebellum

- Second largest part of the brain
- Located behind the brainstem
- Responsible for coordination of voluntary movements and motor control (such as posture and balance)
- Links to frontal and parietal lobes (damage can cause problems with attention, executive functioning, visuospatial ability and emotional control).



Brain Stem

- Connects the brain to the spinal cord.
- Allows messages to travel between the brain and the rest of the body.
- Controls basic bodily functions such as wakefulness, consciousness, heart beat and blood pressure.
- Damage to this area causes lack of consciousness.

Key variables in assessing TBI severity

- Duration of loss of consciousness
- Glasgow Coma Scale Score
- Duration of post traumatic amnesia

Severity of TBI

Mild TBI

- Brief loss of consciousness, usually a few seconds or minutes
- PTA for less than 1 hour of the TBI
- Normal brain imaging results
 - Complicated MTBI – as above plus intracranial abnormalities on imaging

Moderate TBI

- Loss of consciousness for 1 – 24 hours
- PTA for 1 – 24 hours of the TBI
- Abnormal brain imaging results

Severe TBI

- Loss of consciousness or coma for more than 24 hours
- PTA for more than 24 hours of the TBI
- Abnormal brain imaging results

Degree of severity of brain injury

	GCS	PTA	LOC
Mild	13-15	<1 day	0-30 mins
Moderate	9-12	>1 day <7	>30 mins <24 hours
Severe	3-8	>7 days	>24 hours

GCS = Glasgow Coma Scale, PTA = Post Traumatic Amnesia, LOC = Loss of Consciousness

Post traumatic amnesia

- The interval of time between injury and the return of continuous memory for everyday events
- Characterised by confusion and disorientation and severe anterograde amnesia

Effects of TBI

Traumatic Brain Injury

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graph TD; A[Traumatic Brain Injury] --> B[COGNITION]; A --> C[PERSONALITY]; A --> D[PHYSICAL];
```

COGNITION

PERSONALITY

PHYSICAL

Physical Consequences of TBI

- **Paralysis & weakness** – reduced mobility & dexterity, **Fatigability** – tire easily on cognitive tasks (65%)
- **Sensory problems-**
 - **Reduced sense of smell & taste** - (20% SHI)
 - **Vision** –diplopia, blindness, blurred vision
 - **Hearing** - loss – tinnitus
 - **Sensation** - reduced
- **Dizziness/balance problems** - tremor ataxia
- **Speech impairment** –dysphagia
- **Headaches** - pain
- **Epilepsy**
- **Sleep disturbance**
- **Reduced sexual drive**

Cognitive Consequences of TBI

- **Memory –**
 - short term, prospective, new learning
- **Attention & concentration –**
 - sustained, divided, > distractibility
- **Speed of Information Processing – slower**
- **Executive skills impairment –**
 - planning, organising, multi tasking – rigid thinking – synthesising information – problem solving – decision
- **Perceptual and visuo-spatial – neglect**
- **Language –**
 - aphasia – receptive & expressive- word finding
- **Lack of insight & Awareness –** overrates ability, underrates disability

Behavioural Consequences of TBI

- **Disinhibition:** impulsive, reduced behavioural control - poor social judgement – inappropriat beh
- **Lowered tolerance of frustration** – irritability, quick to anger (Orbitofrontal)
- **Apathy and Lack of drive:** poor motivation < initiative -ngug (Dorsolateral)
- **Emotional flattening** - reduced emotional tone, blunting
- **Emotional lability** - easily tearful or laughter - emotional incontinence
- **Egocentricity** - difficulty empathising – put self in others' shoes - inflexibility rigid - self centred
- **Depression & Anxiety** – usually secondary

Difficulties after TBI

(well documented and pessimistic)

- > Risk Psychiatric Disorders (Koponen 2002)
- Axis I Disorders 48% after TBI:
 - Depression - 42 -60%:
 - Alcohol abuse - 12%:
 - Anxiety - 20%:
 - Psychosis 7%
- Axis II Disorders – Personality Disorder - 23%:
- Behavioural problems
- Social isolation
- < Return to work
- > Suicide risk
- > Divorce separation

Life after Head Injury – the most difficult challenge?

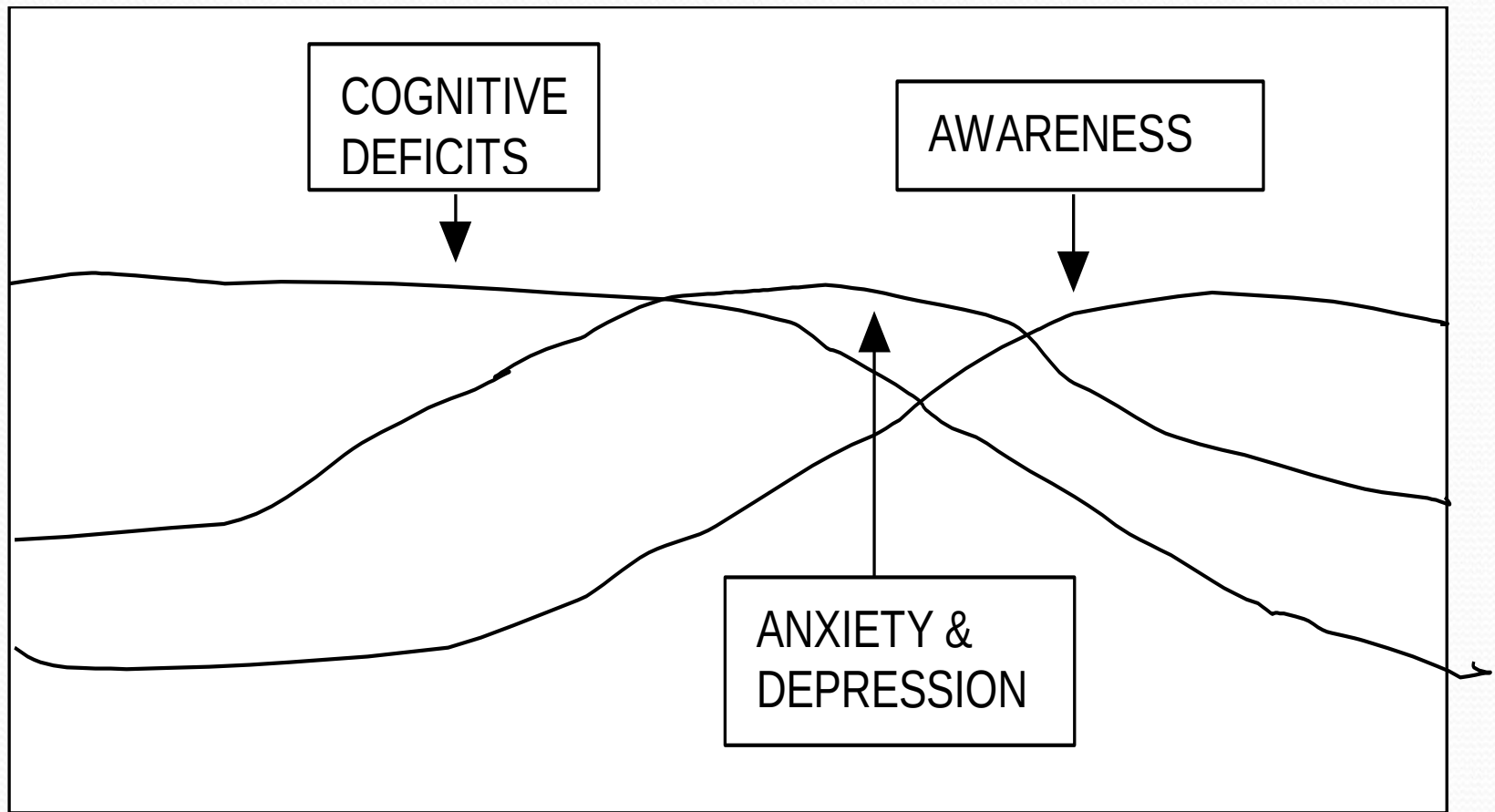
- Within AMH – TBI - “the most difficult challenge”. Most traumatic life event? Threat to life, but also to ‘self’
- Losses - multiple, severe, permanent - physical, cognitive & temperamental – roles, identity, cohesion - bereavement
- It can challenge nearly every aspect of you – walk, talk, think, feel, work, relate, play, past, present and future
- Previous coping strategies often unavailable + initially not aware of problems

Stages of recovery after head injury -

Proposed by Sbordone 1990

- Stage 1 – injury/ coma
- Stage 2 - agitation/confusion disorientation
- Stage 3 – severe cognitive/behavioural deficits – poor insight/ denial
- Stage 4 – moderate cognitive deficits – increased awareness - mild depression
- Stage 5 – significant depression – given little hope of further recovery
- Stage 6 – acceptance of residual deficits

STAGES OF RECOVERY: As cognition improves, awareness increases, and depression may be an inevitable stage - Fordyce (1986)



Bereavement and Loss: The Four Tasks of Mourning

(J. W. Worden)

- To accept the reality of the loss
- To work through the pain of grief
- To adjust to an environment in which the deceased is missing
- To emotionally relocate the deceased and move on with life

Both Restoration & Transformation

(J McGrath)

- Traditional approach to brain injury rehabilitation invokes a model of restoration of the previous person. The 're' prefix in restoration, rehabilitation, recovery, and remediation is backward looking.
- Only partial restoration is possible, which leads to disappointment, grief and pessimism
- If negative outcomes are to be avoided equal emphasis needs to be placed on enabling the development of a new person – transformation – new - change

Shattered Vase: What to do? (1)



Shattered precious vase (2)

- Put it back together exactly as it was with glue and sticky tape?
- Collect the pieces and throw it in the rubbish bin?
- Pick up the beautiful pieces and make something new such as a colourful mosaic?

The big issues and questions

- The big Issues

- Understand the implications of the injury
- To re-establish a sense of meaning in life and the future
- To achieve a sense of self-acceptance and establish a positive self image

- The big Questions

- Why did it happen to me?
- Will I get back to 'normal'? Partial recovery?
- What can I still do and what are my limitations?
- How will I cope?
- What is important in life now?

United States

Lezak (1978)

- Personality changes most likely to create problems for families
1. Impaired social perceptiveness
 2. Impaired control+self-regulation
 3. Stimulus-bound behaviour
 4. Inability to profit from experience
 5. Emotional alterations

“Characterological” Changes After TBI” (Lezak, 1978)

- “...impaired capacity for social perceptiveness”; “...empathy and self-reflective or self-critical attitudes are greatly diminished if not lacking altogether”
- “...impaired capacity for control and self-regulation gives rise to impulsivity, random restlessness, and impatience.”
- “...decreased or absent behavioral initiative...”

“Characterological” Changes After TBI” (Lezak, 1978)

- “apathy, silliness, lability, irritability, and either greatly increased sexual interest or a virtual loss of the sex drive.”
- “...inability to profit from experience compromises the patient’s capacity for social learning....”

Relative's Complaints

- Most difficult problem to live with – *change in personality*

Severe TBI and Personality

- Egocentric/childish
- Communication style-garrulous, tangential
- Impulsive
- Unconcern
- Disinhibited
- Poor temper control
- Impaired judgement
- Insightless
- Aspontaneity
- Loss of drive/apathetic
- Mild euphoria –absence of affect

Damasio

Orbital-frontal	Medial-frontal	Dorso-lateral
poor social judgement	Lacks spontaneity	Poor planning/ organising
shallow affect	lethargic	Distractible
unconcerned	Lack of volition	Concrete
irritable	Placidity	Ritualistic
Over-active/ impulsive	Lack of expression	Perseverative
tactless		
egocentric		
	<i>See: Worthington AD and Wood RLI 2008</i>	

Executive Function: Fractionation of prefrontal functions:

- Orbito-frontal cortex:
 - Implicated in inhibition of learned associations and instigation of new associations between reward and behaviour
 - Network between OFC, sensory areas, amygdala and medial temporal lobe allows for emotionally salient learning.
- Dorsolateral Prefrontal cortex:
 - Manipulation and selection of information from sensory and memory areas; working memory and selective attention
- Medial frontal cortex and anterior cingulate:
 - Initiation of a response and direction of attentional resources towards a controlled behaviour or mental act, especially in face of interference or distraction

Cambridge

Johnson 1989, 1998

Neuropsychological Rehabilitation, 8, 61-79

- 3 and 10 yr FUP
- return to work outcome
- Severe tbi [n=64]
- Pattern at 10 yr essentially the same 18-24m after injury for 85%
- Not work essentially if PTA>6 weeks
- Stresses importance of structured/supported return to work

Johnson 1998: Change in work status at 3 and 10 year follow-ups

3yr/ /10yr	Stable work	Unstable work	No or 1 attempt
Stable work	21	4	0
Unstable work	3	8	0
No or 1 attempt	2	0	24

Phineas Gage



Harlow J.M (1848). Passage of an iron rod through the head.
Boston Med. Surg. J. 39: 389–393

- *‘His contractors, who regarded him as the most efficient and capable foreman in their employ previous to his injury, considered the change in his mind so marked that they could not give him his place again. He is fitful, irreverent, indulging at times in the grossest profanity (which was not previously his custom), manifesting but little deference for his fellows, impatient of restraint of advice when it conflicts with his desires, at times pertinaciously obstinent, yet capricious and vacillating, devising many plans of future operation, which are no sooner arranged than they are abandoned in turn for others appearing more feasible. In this regard, his mind was radically changed, so decidedly that his friends and acquaintances said he was “no longer Gage.”’*



Gage in later life...

- ...from 1851 until just before his death, Gage worked as a stagecoach driver in a livery stable ...and then in Chile for 7 years.
- In 1859, with his health deteriorating, he went to live with his mother and died in San Francisco in 1860 of epileptic convulsions arising as a result of his head injury some 13 years after the accident.



Case 4: James Cracknell

- <http://www.youtube.com/watch?v=gIWj8g88Zaw>
- Checked Nov 14

Most common symptoms

NBFI x DSMIV-criterion A (25-29% cases)

- Tires easily during physical activity
- Easily distracted
- Easily angered or irritated
- Can't get mind off certain thoughts
- Frustrated
- Move slowly

Marital Discord After TBI

- Recent studies show divorce rates ranging from 15% to 54% (Tate et al., 1989; Anderson-Parente, 1990; Wood & Yurdakul, 1997; Webster et al., 1999; Wood et al., 2005; Kreutzer et al., 2007).
- Older persons less likely to divorce (Andersen-Parente, 1990; Kreutzer et al., 2007)
- Persons married longer prior to injury more likely to stay together (Kreutzer et al., 2007).

Role of Neuropsychologist in medico legal work

- Detailed assessment of severity of injury
- Detailed assessment of effects of injury - strengths and weaknesses (physical, cognitive, behavioural)
- Assessment of symptom validity - effort
- Identification of what was caused directly by the accident and secondary problems
- Identification of pre –injury factors that contribute to the problems
- Prognosis – improvement, return to work, effect on quality of life, earning potential
- Therapy, treatment and support

Effort variable in psychometrics

- Effort is the biggest variable in psychometric test results: bigger than other variables such as Brain Damage and Anxiety or PTSD.
- Paul Green: 'The power of the external incentive'. WMT failures 23 times higher in MTB than parents seeking custody

Poor effort and deception is common

- In medico-legal work claimants exaggerate and some frankly lie. Many studies suggest that 20-30% feign impairment.
- But
- Bunnage (2008) In routine NHS outpatient neuropsychology clinic out of 132 consecutive referrals, 30% failed the WMT. 9% had potentially external factors. Out of that 9% 50% failed the WMT

Potential factors and incentives relating to exaggeration of symptoms

- Entitlement/justification/anger etc.
- Exaggeration to prove a point
- Somatisation/Fictitious/conversion/unconscious
- Misattribution/ Reinforced behaviour/
- Malingering for specific financial/external gain
- Avoid responsibility/criminal
- Attention from spouse/parent/doctor
- Having a bad day

Posttraumatic growth - What is it? An old idea – growth through adversity.

- Religion: Christianity, Buddhism, Islam, philosophy, literature, Greek tragedies. Meaning and change emerge from the struggle with tragedy
- Positive psychology research: 40-70% of people who experience trauma report some form of benefit.
- “What doesn’t kill you makes you stronger” Nietzsche. “The strongest steel goes through the hottest fire”. Tragedy springboard for transformation.

First Study: Powell et al (2007) Posttraumatic growth after Head Injury: a long term follow up Brain Inj;21:1335-1351

- Comparing two groups from Reading Head Injury Clinic. 1.7 years (2003 - 4) and 11.6 years (1993 - 4) post injury,
- Postal study - given a battery of questionnaires –.
- 24 people in each group
- Measures: Severity of injury, Disability, Life satisfaction, Anxiety, Depression, Posttraumatic growth

First Study Result

- No significant difference on questionnaires for: Severity of Injury, Disability, Life Satisfaction, Anxiety, Depression, but ...
- Significant difference on The Posttraumatic Growth Inventory . 11.6 years post injury score on PTGI = 68: 1.7 years post injury score on PTGI = 36 ($p < 0.000$)
- Light at the end of the tunnel. Early group still in middle of it – something happens - change or growth – silver lining?

Second Posttraumatic Growth

study: (Powell et al 2012: Disability and Rehabilitation;34(16-17):1461-1467

- What are the factors that are associated with posttraumatic growth 13 years post injury?
- Why do some people experience high posttraumatic growth and others do not?

Study 2: What factors might influence Posttraumatic Growth?

- Structured interviews and questionnaire study of people approx 13 years post injury. Interviewed 21 people
- Would they score similarly on the posttraumatic growth questionnaire 2 years after first completing it?
- What variables are associated, correlate or predict with posttraumatic growth?

Factors associated with PTG 13 years post TBI

- Having a sense of purpose and meaning
- Having activities & life events happen
- Social support network – quality of relation
- Work
- A less severe physical disability
- Religiousness
- Having a new partner or partner
- Optimism

Rehabilitation is a long journey and recovery is invariably partial



Thank You

